Relations and Functions

- **Definition 1**
  Let X be a set. A **relation** $\sim$ on X is a rule that determines for every $x, y \in X$ whether the statement
  $$x \sim y$$
  is true or false. If it is true, we say that x is related to y. If it is false, we write $x \nsim y$.
    - **Example 1.1**
      Let $X = \{1, 2, 3, 4\}$, the following are examples of relations on X.
        1. The relation $\sim$ given by $1 \sim 2, 2 \sim 3, 3 \sim 1, 4 \sim 4$.
        2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 2 \sim 4$.
        3. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
- **Definition 2**
  Let X be a set and $\sim$ a relation on X.
    1. The relation is **reflexive** if $x \sim x$ for every $x \in X$.
    2. The relation is **symmetric** if for every $x, y \in X$
       $$x \sim y \implies y \sim x.$$
    3. The relation is **antisymmetric** if for every $x, y \in X$
       $$x \sim y \wedge y \sim x \implies x = y.$$
    4. The relation is **transitive** if for every $x, y, z \in X$
       $$x \sim y \wedge y \sim z \implies x \sim z.$$
    - **Example 2.1**
      On the set $\mathbb{Z}$ consider the relation
      $$x \sim y \iff x \leqslant y.$$
      Note that
        1. $\sim$ is reflexive since $x \leqslant x$ for every $x \in \mathbb{Z}$.
        2. $\sim$ is not symmetric since $1 \leqslant 2$ but $2 \nleqslant 1$.
        3. $\sim$ is antisymmetric because if $x \leqslant y$ and $y \leqslant x$ then $x = y$.
        4. $\sim$ is transitive since $x \leqslant y$ and $y \leqslant z$ implies that $x \leqslant z$.
- **Definition 3**
  Let X be a set. An **equivalence relation** on X is a relation $\sim$ that satisfies the following properties.
    1. $\sim$ is reflexive.
    2. $\sim$ is symmetric.
    3. $\sim$ is transitive.
    - **Example 3.1**
      Let $X = \{1, 2, 3, 4\}$. The following are examples of equivalence relations on X.
        1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
        2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 2 \sim 3, 3 \sim 2, 1 \sim 4, 4 \sim 1$.
- **Definition 4**
  Let X be a set. An **order relation** on X is a relation $\sim$ that satisfies the following properties.

1. $\sim$ is reflexive.
2. $\sim$ is antisymmetric.
3. $\sim$ is transitive.

  - **Example 4.1**
    Let $X = \{1, 2, 3, 4\}$. The following are examples of order relations on X.
    1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.
    2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 1 \sim 2, 1 \sim 3, 1 \sim 4, 2 \sim 3, 2 \sim 4, 3 \sim 4$.

- **Definition 5**
  A **function** $f\colon X \to Y$ is a correspondence of elements of X with elements of Y such that forall $x \in X$ there exists a unique $y \in Y$ such that $y = f(x)$.

$$\begin{aligned} f\colon X &\to \ Y \\ x &\mapsto \ y = f(x) \end{aligned}$$

  The element $y = f(x)$ is called the **image** of x under f and the element x is called the **preimage** of y under f.

- **Definition 6**
  If $f\colon X \to Y$ is a function, the **domain** of f denote $\mathrm{Dom}(f)$ is X and the **codomain** of $\mathrm{CoDom}(f)$ is Y. The **range** of f, denoted $\mathrm{Ran}(f)$, is the set

$$\mathrm{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$$

  Note that $\mathrm{Ran}(f) \subseteq Y$.

  - **Example 6.1**
    1. The function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
    2. The function $g\colon \mathbb{R} \to \mathbb{R}$, $g(x) = \sin(x)$.
    3. The function $h\colon [0, \infty) \to \mathbb{R}$, $h(x) = \sqrt{x}$.

  - **Example 6.2**
    Let $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$ the following examples of functions from X to Y.
    1. The function $f\colon X \to Y$ defined by
    $$f(1) = a \quad f(2) = b \quad f(3) = c.$$
    2. The function $g\colon X \to Y$ defined by
    $$g(1) = a \quad g(2) = a \quad g(3) = a.$$
    3. The function $h\colon X \to Y$ defined by
    $$h(1) = a \quad h(2) = b \quad h(3) = a.$$

- **Definition 7**
  Two functions $f\colon X \to Y$ and $g\colon Z \to W$ are **equal** if $X = Z, Y = W$ and $f(x) = g(x)$ for all $x \in X$.

  - **Example 7.1**
    1. Let X be a set. The **identity function** on X is the function $\mathrm{id}_x\colon X \to X$ given by $\mathrm{id}_x(x) = x$.
    2. Let X and Y be sets and $c \in Y$. The **constant function** with value c is the function $\mathrm{K}_c\colon X \to Y$ given by $\mathrm{K}_c(x) = c$.
    3. Let X be a set and $A \subseteq X$. The **inclusion function** of A in X is the function $\iota\colon A \to X$ (or $\iota\colon A \hookrightarrow X$) given by $\forall x \in A, \iota(x) = x$.

4. If $f\colon X \to Y$ is a function and $A \subseteq X$, the **restriction** of f to A is the function $f\big|_A\colon A \to Y$ given by $f\big|_A(x) = f(x)$ for all $x \in A$.

- **Definition 8**
  If $f\colon X \to Y$ and $g\colon Z \to W$ are functions, the **composition** of g with f, denoted $g \circ f$, is the function $g \circ f\colon X \to Z$ defined by $(g \circ f) = g\left(f\left(x\right)\right)$ for all $x \in X$.
  - **Example 8.1**
    Note that if $f\colon X \to Y$, f is a function and $A \subseteq X$, we have that $f\big|_A = f \circ \iota_A$ where $\iota\colon A \to x$.

$$A \xrightarrow{\iota} X \xrightarrow{f} Y.$$

- **Remark 8**
  Note that even in the case both $g \circ f$ and $f \circ g$ are defined, the composition of two functions is not commutative, that is, in general $g \circ f \neq f \circ g$.
  - **Example 8.2**
    Consider the functions

$$\begin{aligned} f{:}\mathbb{R} &\to \mathbb{R}, & g\colon \mathbb{R} &\to \mathbb{R}. \\ x &\mapsto x^2, & x &\mapsto x+1. \end{aligned}$$

    Note thet $g \circ f\colon \mathbb{R} \to \mathbb{R}$ and $f \circ g\colon \mathbb{R} \to \mathbb{R}$ with

$$\begin{aligned} g \circ f &= g\left(f\left(x\right)\right) & f \circ g &= f\left(g\left(x\right)\right) \\ &= g\left(x^2\right) & &= f\left(x+1\right) \\ &= x^2+1 & &= \left(x+1\right)^2. \end{aligned}$$

    In particular if $x = 2$ we have that $(g \circ f)(2) = 5$ and $(f \circ g) = 9$. Therefore $g \circ f \neq f \circ g$.
  - **Proposition 8.3**
    Let $f\colon X \to Y$, $g\colon Y \to Z$ and $h\colon Z \to W$ be functions, the $(h \circ g) \circ f = h \circ (g \circ f)$. In other words, the composition is associative.
    - **Proof 8.3.1**
      Let $x \in X$ then

$$\begin{aligned} [(h \circ g) \circ f](x) &= (h \circ g)(f(x)) \\ &= h(g(f(x))) \\ &= h((g \circ f)(x)) \\ &= [h \circ (g \circ f)](x). \end{aligned}$$

      Now, as $[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x)$ for all $x \in X$ we have that $(h \circ g) \circ f = h \circ (g \circ f)$.
  - **Example 8.4**
    Let $s, t\colon \mathbb{Z} \to \mathbb{Z}$ be functions defined by $s(x) = x+1$ and $t(x) = 2x$. Prove that $t \circ s \neq s \circ t$.
    - **Proof 8.4.1**
      Let $x \in \mathbb{Z}$ then

$$\begin{aligned} t \circ s &= t\left(s\left(x\right)\right) & s \circ t &= s\left(t\left(x\right)\right) \\ &= t\left(x+1\right) & &= s\left(2x\right) \\ &= 2x+2 & &= 2x+1. \end{aligned}$$

      In particular if $x = 2$ we have that $(t \circ s)(2) = 6$ and $(s \circ t)(2) = 5$. Therefore $t \circ s \neq s \circ t$.
  - **Example 8.5**
    Let $X = \{1, 2, 3, 4, 5\}$ and let $f\colon X \to X$ be the function defined by

$$f(1)=2, \quad f(2)=2, \quad f(3)=4, \quad f(4)=4, \quad f(5)=4.$$

Show that $f \circ f = f$. Find a function g such that $g \circ f = f$ and $f \circ g = f$.

- **Solution 8.5.1**

$$g: X \to X$$
$$g(1)=1, \quad g(2)=2, \quad g(3)=5, \quad g(4)=4, \quad g(5)=3.$$

Then

$$\begin{aligned}
(g \circ f)(1) &= g(f(1)) = g(2) = 2 = f(1), \\
(g \circ f)(2) &= g(f(2)) = g(2) = 2 = f(2), \\
(g \circ f)(3) &= g(f(3)) = g(4) = 4 = f(3), \\
(g \circ f)(4) &= g(f(4)) = g(4) = 4 = f(4), \\
(g \circ f)(5) &= g(f(5)) = g(4) = 4 = f(5).
\end{aligned}$$

Therefore $g \circ f = f$. Similarly, we can prove $f \circ g = f$ and $f \circ f = f$.

- **Definition 9**
  Let $f: X \to Y$ be a function.
  We say that f is **injective** if given $x_1, x_2 \in X$, such that $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
  Equivalently, if given $x_1, x_2 \in X$ such that $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.

  We say that f is **surjective** if for every $y \in Y$ there exists $x \in X$ such that $y = f(x)$.

  We say that f is **bijective** if it is both injective and surjective.

- **Remark 9**
  If there exists a bijective function $f: X \to Y$. We say that X and Y are **one-to-one correspondence**.

  - **Example 9.1**
    The identity function $\mathrm{id}_x: X \to X$ is injective since if $\mathrm{id}_x(x) = \mathrm{id}_x(y)$ then $x = y$.
    It is also surjective since for all $x \in X$ we have that $x = \mathrm{id}_x(x)$.
    Therefore, id_x is bijective.

  - **Example 9.2**
    Consider the constant function $\mathrm{K}_c: X \to Y$.
    Note that if X has more than one element the function is not injective and if Y has more than one element the function is not surjective.

  - **Example 9.3**
    The inclusion function $\iota: A \to X$ is injective since $\iota_A(x) = \iota_A(y)$ implies $x = y$. Note that, if $A = X$ then $\iota_x = \mathrm{id}_x$. Therefore ι_A is surjective. If $A \neq X$ and is not surjective if $A \subset X$.

  - **Example 9.4**
    Composition and Injectivity/Surjectivity
    Suppose that $f: A \to B$ and $g: B \to C$ are functions. Prove that if $g \circ f$ is injective then f is injective, and if $g \circ f$ is surjective then g is surjective.

    - **Solution 9.4.1**
      Proof:

      - **First Part (If $g \circ f$ is injective, then f is injective):**
        Let $x_1, x_2 \in A$ be arbitrary elements in the domain of f such that:

        $$f(x_1) = f(x_2)$$

        Since g is a function, applying g to both sides yields identical outputs:

        $$g(f(x_1)) = g(f(x_2))$$

        By definition of function composition, this is equivalent to:

$$(g \circ f)(x_1) = (g \circ f)(x_2)$$

We are given the hypothesis that $g \circ f$ is injective. By definition of an injective function, $(g \circ f)(x_1) = (g \circ f)(x_2) \implies x_1 = x_2$.
Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$, proving that f is injective.

- **Second Part (If $g \circ f$ is surjective, then g is surjective):**
  Let $z \in C$ be an arbitrary element in the codomain of g.
  We are given the hypothesis that $g \circ f: A \to C$ is surjective. By definition of a surjective function, there exists some element $x \in A$ such that:

  $$(g \circ f)(x) = z$$

  By definition of function composition, this means:

  $$g(f(x)) = z$$

  Let $y = f(x)$. Since $x \in A$ and $f: A \to B$, it follows that $y \in B$.
  Substituting y back into the equation gives $g(y) = z$.
  Thus, for any arbitrary $z \in C$, there exists an element $y \in B$ such that $g(y) = z$, proving that g is surjective.